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WinALDL screenshots

Functionallity of the program

At the top of the program you have three buttons. One opens the data logging dialog, one opens the configuration dialog and the last is the EXIT button. These are allways visable. The main part of the program is used to display engine data. Click on the tabs to select what data to dispaly.

Configuration

The program is configured using a configuration dialog.

The Configuration dialog box contains the following fields and controls:

- ECM Type: A dropdown menu showing '1227747' and an 'EXIT' button.
- COM Port: A dropdown menu showing 'COM1' and a checkbox labeled 'Restart program after changing COM Port'.
- TPS Voltage for 0%: A text box with '0.54', a spin box with '0.54', and a range '0.00 - 2.00 (default = 0.54)'.
- TPS Voltage for 100%: A text box with '4.60', a spin box with '4.60', and a range '2.01 - 5.00 (default = 4.60)'.
- RPM range for 'Narrow': A text box with '80', a spin box with '80', and a range '(default = 80)'.
- MAP range for 'Narrow': A text box with '1.99', a spin box with '1.99', and a range '(default = 2.00)'.

All configuration changes takes imideate effect exept for the com port. The configuration is stored and remebered next time WinALDL is started. If the com port is changed the program must be restarted.

Data logging

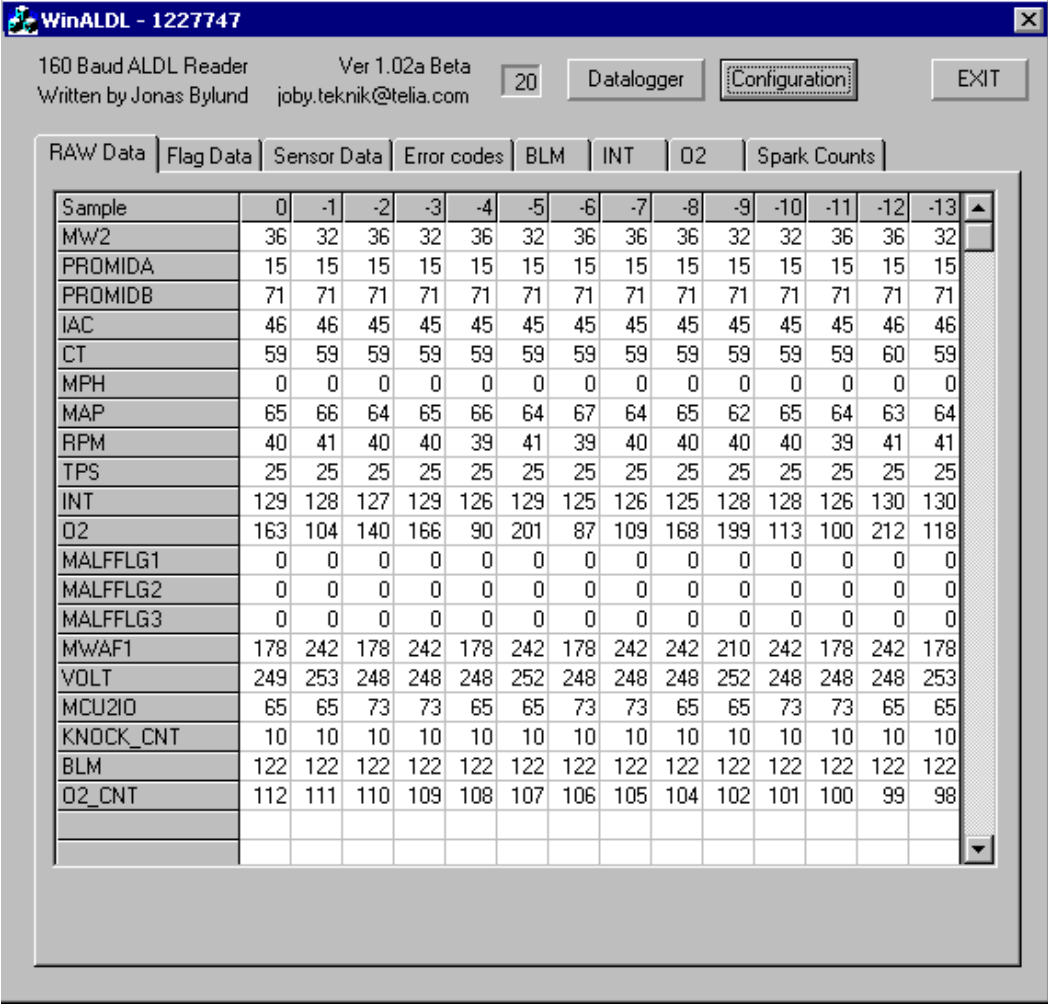
This is the data logging dialog. You can select what data you want to log. The only limit for how long you can log is the available disk space.

The LOG Data dialog box contains the following controls:

- Checkboxes: ☐ RAW Data, ☒ Flag Data, ☒ Sensor Data, ☐ Error codes.
- Radio buttons: ☒ US, ☐ Metric.
- Buttons: 'EXIT' and 'START'.

RAW Data

This tab shows a grid with raw ALDL data in decimal form. The data is scrolled from left to right as new data are read.



WinALDL - 1227747

160 Baud ALDL Reader Ver 1.02a Beta
Written by Jonas Bylund joby.teknik@telia.com

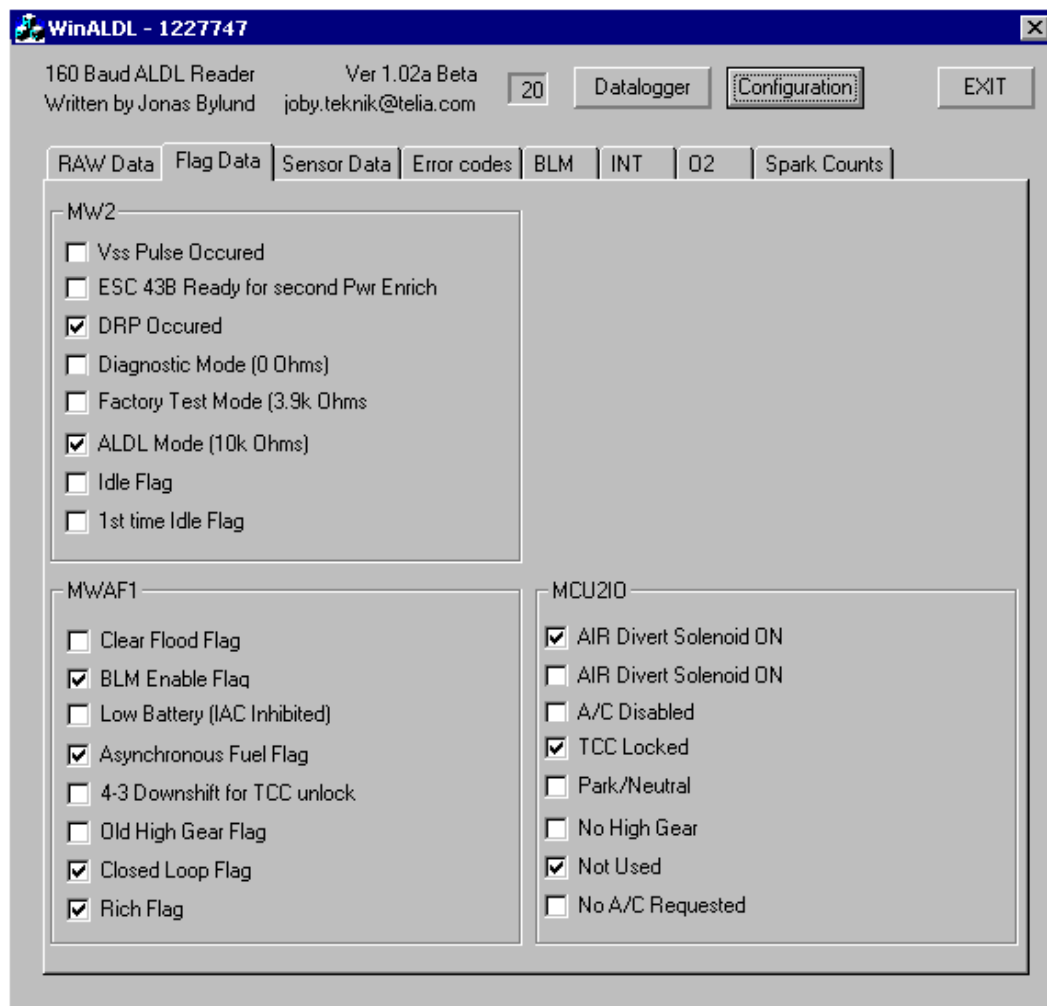
20 Datalogger Configuration EXIT

RAW Data Flag Data Sensor Data Error codes BLM INT O2 Spark Counts

Sample	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
MW2	36	32	36	32	36	32	36	36	36	32	32	36	36	32
PROMIDA	15	15	15	15	15	15	15	15	15	15	15	15	15	15
PROMIDB	71	71	71	71	71	71	71	71	71	71	71	71	71	71
IAC	46	46	45	45	45	45	45	45	45	45	45	45	46	46
CT	59	59	59	59	59	59	59	59	59	59	59	59	60	59
MPH	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MAP	65	66	64	65	66	64	67	64	65	62	65	64	63	64
RPM	40	41	40	40	39	41	39	40	40	40	40	39	41	41
TPS	25	25	25	25	25	25	25	25	25	25	25	25	25	25
INT	129	128	127	129	126	129	125	126	125	128	128	126	130	130
O2	163	104	140	166	90	201	87	109	168	199	113	100	212	118
MALFFLG1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MALFFLG2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MALFFLG3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MWAF1	178	242	178	242	178	242	178	242	242	210	242	178	242	178
VOLT	249	253	248	248	248	252	248	248	248	252	248	248	248	253
MCU2IO	65	65	73	73	65	65	73	73	65	65	73	73	65	65
KNOCK_CNT	10	10	10	10	10	10	10	10	10	10	10	10	10	10
BLM	122	122	122	122	122	122	122	122	122	122	122	122	122	122
O2_CNT	112	111	110	109	108	107	106	105	104	102	101	100	99	98

Flag Data

This tab shows a lot of check boxes that indicate if flag data are true or false. Here you can see what mode the ECM is operating in. You can also see the status of digital input and digital output.



Sensor Data

This tab shows analog engine data. Both raw values and converted. At the bottom you also have a spin button that is used to select one of three modes of operation. Latest values, Max values or Min values. You also have a clear button to clear Max / Min values.

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Ver 1.02a Beta
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Datalogger

Configuration

EXIT

RAW Data
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	RAW	Converted	Unit	Converted	Unit
IAC	46	46	[#]		
Coolant Temp	60	177.1	[F]	80.6	[C]
MPH	0	0	[MPH]	0	[KPH]
MAP	63	1.2	[Volts]	33.6	[kpa]
RPM	40	1000	[RPM]		
TPS	25	0.5	[Volts]	9.8	[%]
INT	128	128	[#]		
O2	190	0.840	[Volts]		
Battery Voltage	129	12.9	[Volts]		
Knock counter	10	10	[#]		
BLM	123	123	[#]		
rich / lean counter	18	18	[#]		

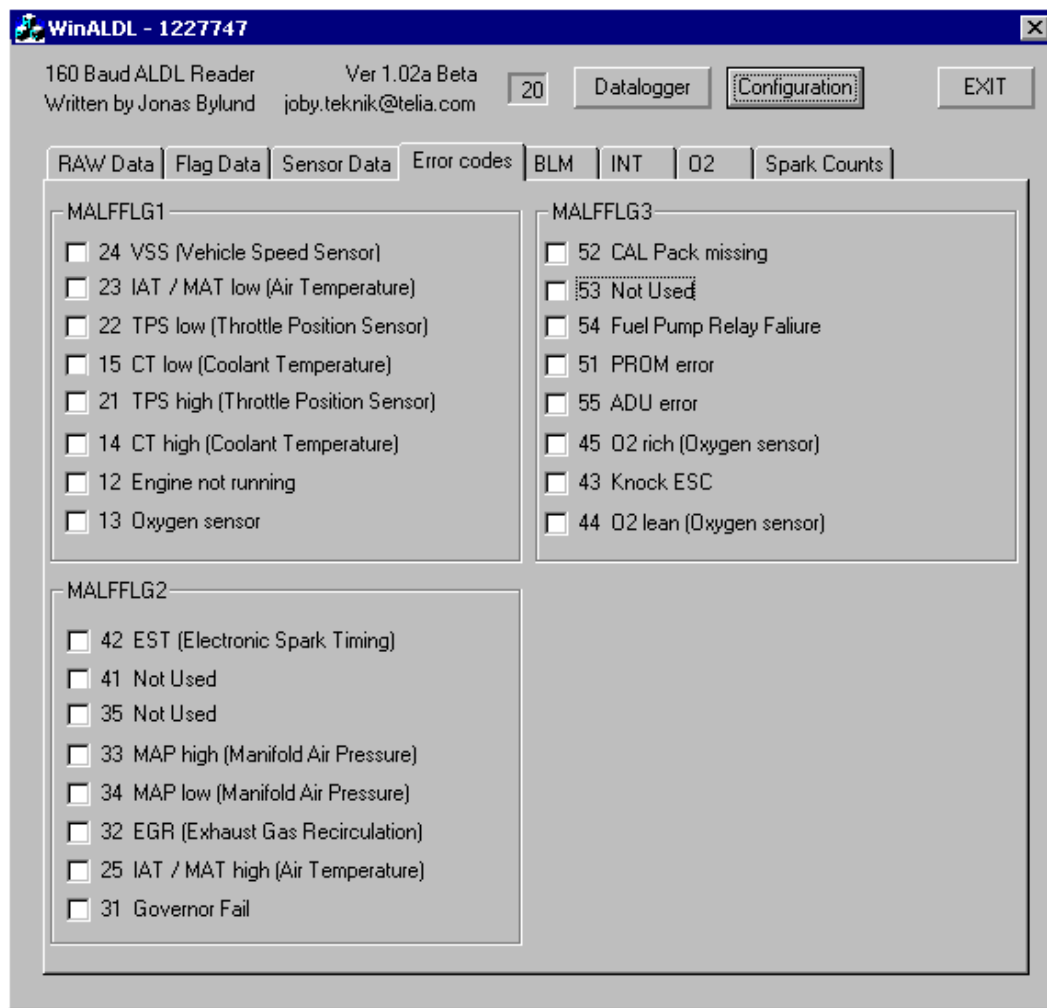
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▼

Latest values

Error codes

Here you see all error codes. Check boxes are used to indicate if the error code is set or not.



BLM, INT and O2

These three tabs all look the same. It is a grid with RPM and MAP axes. When the program is started all cells are empty. As you run the engine and the engine operating point travels over the chart the cells will start to populate with engine data. O2 will start to populate right away, but INT and BLM starts when the engine is in closed loop. At the bottom you have one button to clear the table to start over and one button to save the current table values. The table will be stored in 'My documents' directory, and a dialog box will tell you exact location and name of the file. At the bottom you also have a spin button that is used to select one of four modes of operation:

Narrow Latest - Discard all data that is not close to the axes values. Show latest data in each cell.

Narrow Avg - Discard all data that is not close to the axes values. Show averaged data in each cell.

Narrow # Samples - No of samples in each cell in the 'Narrow Avg' selection.

Narrow Avg 10 - Average of the 10 latest samples.

Narrow Std. Dev. 10 - Standard deviation for the 10 latest samples.

Wide Latest - Do not discard any data. Show latest data in each cell.

Wide Avg - Do not discard any data. Show averaged data in each cell.

Wide # Samples - No of samples in each cell in the 'Wide Avg' selection.

Wide Avg 10 - Average of the 10 latest samples.

Wide Std. Dev. 10 - Standard deviation for the 10 latest samples.

When you save INT / BLM data the file will have two additional tables:

Narrow Correction 10 - Correction factors for the EPROM table based on the values in the 'Narrow Avg 10'. Multiply the cell you want to change in the EPROM with the value in the corresponding 'Narrow Correction 10' cell. The value in the 'Narrow Std. Dev. 10' table is used to see if the data in that cell is good enough to use for a new calibration or not.

Wide Correction 10 - Correction factors for the EPROM table based on the values in the 'Wide Avg 10'. Multiply the cell

you want to change in the EPROM with the value in the corresponding 'Wide Correction 10' cell. The value in the 'Wide Std. Dev. 10' table is used to see if the data in that cell is good enough to use for a new calibration or not.

WinALDL - 1227747 [X]

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20 Datalogger Configuration EXIT

RAW Data Flag Data Sensor Data Error codes BLM INT O2 Spark Counts

RPM \ MAP	20	30	40	50	60	70	80	90	100
400									
800		125.0	125.5	133.0					
1200		126.9	125.7	126.7					
1600		129.3	137.5	144.3	129.7	133.0			
2000	115.0	125.3	123.3	130.5	128.0		130.0		
2400	119.0			126.5	122.0	126.0			
2800	129.7	130.0	133.5	126.0					
3200	129.0			125.0			128.0	128.0	

Clear Table Save Table Wide Avg

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RAW Data

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BLM

INT

O2

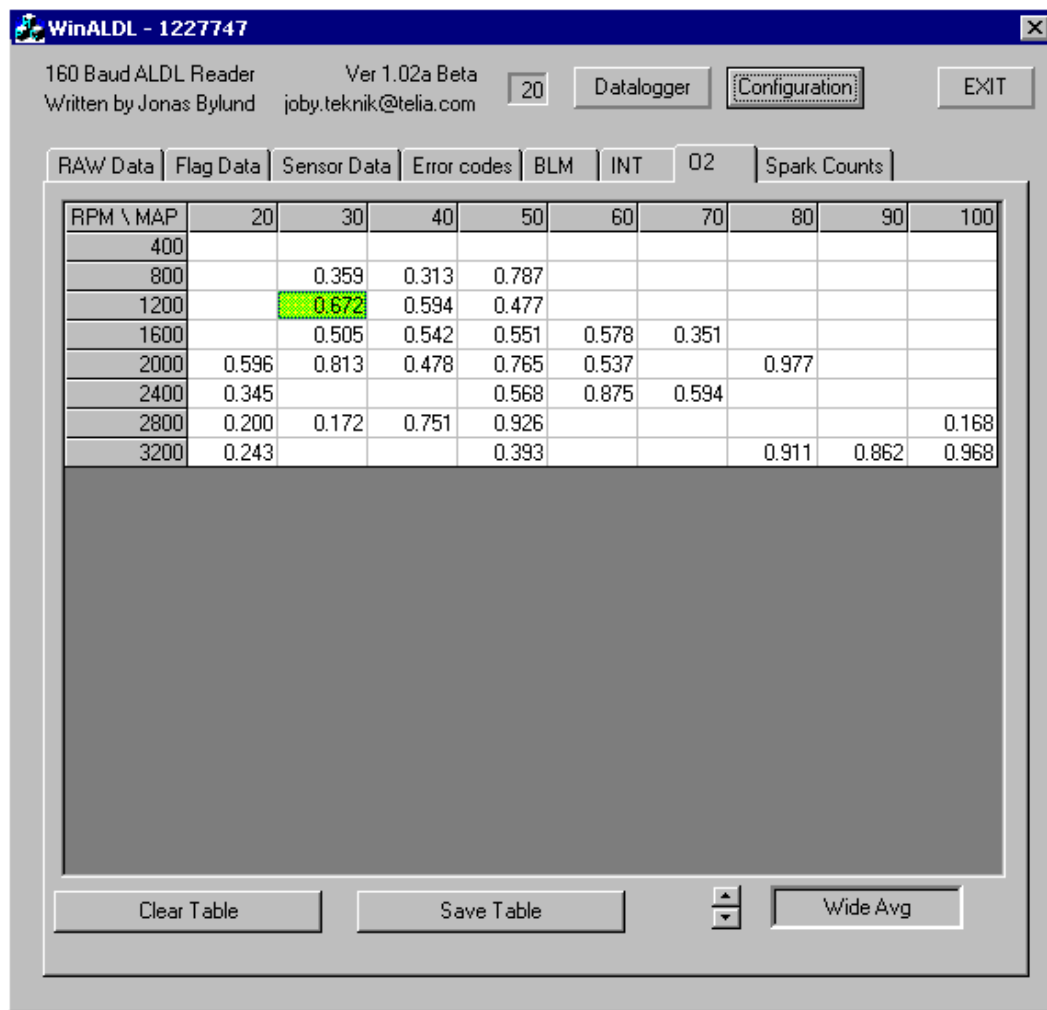
Spark Counts

RPM \ MAP	20	30	40	50	60	70	80	90	100
400									
800		124.2	124.4	130.0					
1200		125.5	126.5	124.0					
1600		130.9	126.7	127.8	123.7	124.5			
2000	130.3	134.0	126.3	131.0	124.5		133.0		
2400	127.0			124.5	125.0	127.0			
2800	128.0	129.0	125.5	127.0					
3200	128.0			125.0			124.0	125.0	

Clear Table

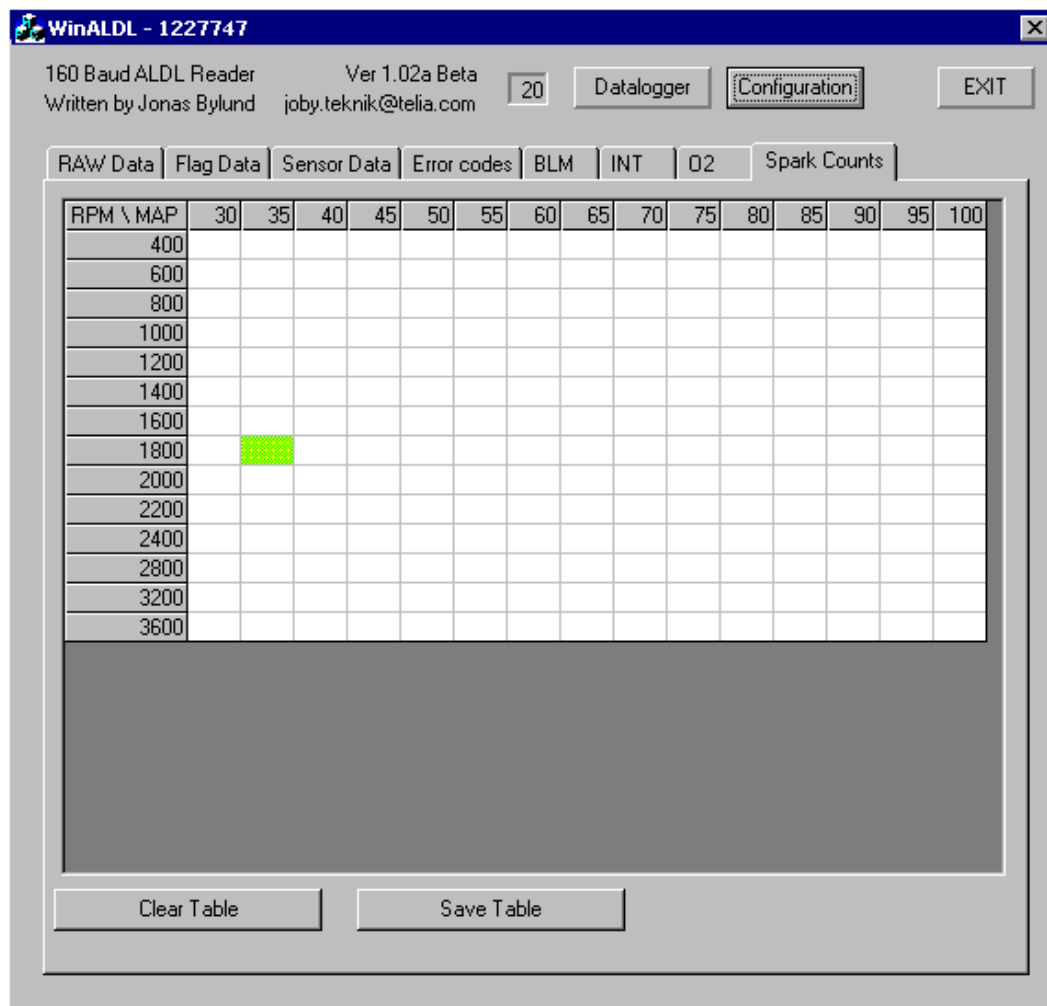
Save Table

Wide Avg



Spark Counts

This is a grid with RPM and MAP axes. When the program is started all cells are empty. Whenever an engine knock is detected the value in the cell with that engine operating point will increase with one. This is used to detect areas where the engine has a lot of knocks. At the bottom you have one button to clear the table to start over and one button to save the current table values. The table will be stored in 'My documents' directory, and a dialog box will tell you exact location and name of the file.



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